Evaluate yourself – 4

H_3BO_3 accepts hydroxide ion from water as shown below

$H_3BO_3(aq)+H_2O(l) \rightleftharpoons B(OH)_4^- + H^+$

Predict the nature of H_3BO_3 using Lewis concept

8.2 Strength of Acids and Bases

The strength of acids and bases can be determined by the concentration of H_3O^+ (or) OH^- produced per mole of the substance dissolved in H_2O. Generally we classify the acids / bases either as strong or weak. A strong acid is the one that is almost completely dissociated in water while a weak acid is only partially dissociated in water.

Let us quantitatively define the strength of an acid (HA) by considering the following general equilibrium.

$$\underset{\text{acid 1}}{HA} + \underset{\text{base 2}}{H_2O} \rightleftharpoons \underset{\text{acid 2}}{H_3O^+} + \underset{\text{base 1}}{A^-}$$

The equilibrium constant for the above ionisation is given by the following expression

$$K=\frac{[H_3O^+][A^-]}{[HA][H_2O]} \qquad .....(8.1)$$

We can omit the concentration of H_2O in the above expression since it is present in large excess and essentially unchanged.

$$K_a=\frac{[H_3O^+][A^-]}{[HA]} \qquad .....(8.2)$$

Here, K_a is called the ionisation constant or dissociation constant of the acid. It measures the strength of an acid. Acids such as HCl, HNO_3 etc... are almost completely ionised and hence they have high K_a value (K_a for HCl at 25°C is 2×10^6) Acids such as formic acid ($K_a=1.8\times10^{-4}$ at 25°C), acetic acid (1.8×10^{-5} at 25°C) etc.. are partially ionised in solution and in such cases, there is an equilibrium between the unionised acid molecules and their dissociated ions. Generally, acids with K_a value greater than ten are considered as strong acids and less than one are considered as weak acids.

Let us consider the dissociation of HCl in aqueous solution,

$$\underset{\text{acid 1}}{HCl} + \underset{\text{base 2}}{H - OH} \rightleftharpoons \underset{\text{acid 2}}{H_3O^+} + \underset{\text{base 1}}{Cl^-}$$

As discussed earlier, due to the complete dissociation, the equilibrium lies almost 100% to the right. i.e., the Cl^- ion has only a negligible tendency to accept a proton form H_3O^+ . It means that the conjugate base of a strong acid is a weak base and vice versa.

The following table illustrates the relative strength of conjugate acid – base pairs.

XII U8-Ionic equilibrium.indd 6 16-12-2022 19:33:29